

Agreement on Partial Domains: Canonical Abstraction and a Δ -System Criterion

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Abstract

Systems with partial observations often compare two states by requiring equal answers only to queries available in both. This common-domain relation is reflexive and symmetric but can fail transitivity, preventing a canonical quotient and permitting incomparable maximal coarsenings. We formalize two repairs in Lean. Treating query availability as part of future behavior yields an equivalence relation and a canonical quotient characterized by observation-kernel refinement. If availability is intentionally ignored, we characterize the domain families that restore transitivity uniformly over arbitrary proposition-valued decision profiles: precisely the Δ -systems, whose distinct domains share one common root and otherwise have private petals. A mediator-hole labeling proves sharpness; explicit fixtures show that equal domains are unnecessary, nested domains and nonempty overlap are insufficient, and fixed labelings may be transitive accidentally outside the Δ -system regime. The contribution is a machine-checked synthesis and design boundary built from classical ingredients, not a claim that partial-function compatibility or Δ -systems are new.

1. Introduction

The paper number places this self-contained note in the broader Δ t Framework corpus, a series on formal boundaries of observation, admissibility, and composition. No result from Papers 1–27 is assumed here.

Many systems compare objects that do not expose the same information. A monitor may have access to some predicates but not others. A policy engine may evaluate only the clauses enabled for one request. A partial specification may leave some input/output behavior undefined. A record may carry only a subset of a schema. In each setting, a natural comparison is:

two objects are compatible when they agree everywhere both are defined.

This comparison is useful and well established. It is also not generally an equivalence relation. The failure is small enough to miss in an interface and large enough to invalidate canonical quotienting.

Suppose states a and c can both answer a query w , while state b cannot. If a and c disagree on w , the disagreement is invisible when either is compared with b . It is therefore possible to have

$$aCb, \quad bCc, \quad \neg aCc.$$

Compatibility is reflexive and symmetric, but the mediator b hides an endpoint disagreement by lacking the query that exposes it. Grouping states by C is not therefore a quotient construction.

This paper gives a precise account of two repairs.

The first is semantic: availability is included in what must be preserved. Two states are equivalent only when they admit the same finite continuations and make the same governed decision after every admitted continuation. The resulting future relation is an equivalence and supports a canonical quotient. An observation is future-sufficient exactly when its kernel refines that quotient kernel.

The second repair retains the weaker semantics—availability differences remain ignored—but constrains the geometry of availability domains. The relevant condition is that every pair of distinct domains share one common root. Outside the root, each state may have a private petal disjoint from every other petal:

$$A_s = K \cup P_s, \quad P_s \cap P_t = \emptyset \quad (s \neq t).$$

This is a Δ -system/common-root family. In the static profile abstraction formalized here, it is exactly the domain-only condition under which common-domain compatibility is transitive for every proposition-valued decision assignment.

The universal quantifier is essential. A fixed assignment can accidentally make compatibility transitive even when the domain geometry is bad; constant decisions are the simplest example. The theorem characterizes a structural guarantee supplied by domains alone, not a necessary condition for every particular system.

The formal contribution has five parts:

1. a common-domain compatibility relation and finite nontransitivity witnesses;
2. a characterization of compatible observation kernels and the distinction between maximal and greatest compatible equivalences;
3. an availability-sensitive future equivalence with a canonical quotient;
4. an equivalence between domain mediation, common-root geometry, and uniform transitivity of static profiles;
5. fixtures proving strictness, sharpness, and the fixed-versus-uniform boundary.

The mathematics uses classical objects. Partial functions agreeing on their common domains, tolerance relations, compatible states in incomplete machines, partial words, and Δ -systems all have substantial prior literatures [1–8]. The claim here is narrower: these ingredients form a useful, machine-checked architecture theorem about when a partial-observation comparison may be treated as canonical.

2. Partial profiles and common-domain compatibility

Let S be a type of states and W a type of finite continuations or queries. For each state $s \in S$, let

$$A_s \subseteq W$$

be its admissibility domain. Let a decision profile be a proposition-valued function

$$d : S \times W \rightarrow \text{Prop}.$$

Only the values $d(s, w)$ with $w \in A_s$ are semantically exposed. Writing $d_s(w)$ for $d(s, w)$, define common-domain compatibility by

$$s C_d t \iff \forall w \in A_s \cap A_t, (d_s(w) \leftrightarrow d_t(w)).$$

In the Lean source, the static definition is `ProfileCompatible` admitted `decision left right`, with `admitted : S → W → Prop` and `decision : S → W → Prop`. The proposition-valued presentation avoids adding decidability hypotheses to the structural theorem. Boolean partial functions provide the immediate finite intuition. The native finite-word continuation model later uses the corresponding relation `CommonCompatible`; the static profile isolates the arbitrary-labeling question used in the uniform characterization.

For every fixed d , C_d is reflexive and symmetric. It need not be transitive. This makes it a tolerance or compatibility relation in the broad relation-theoretic sense [2,3], not automatically an equivalence relation.

2.1 Observation kernels

An observation $O : S \rightarrow Y$ induces the kernel

$$\ker O = \{(s, t) \mid O(s) = O(t)\}.$$

The kernel is always an equivalence relation. The observation is sufficient for the weaker partial-future semantics exactly when

$$O(s) = O(t) \implies sC_d t,$$

or equivalently

$$\ker O \subseteq C_d.$$

Thus admissible observation kernels are equivalence subrelations contained in a relation that may not itself be an equivalence.

This distinction generates two order-theoretic notions that must not be conflated.

- A **maximal** compatible equivalence has no strictly larger compatible equivalence above it.
- A **greatest** compatible equivalence contains every compatible equivalence.

Finite systems always have maximal compatible equivalences: the identity relation is compatible, and a finite inclusion order has maximal elements. They need not have a greatest one. The formal three-state fixture admits two distinct incomparable maximal choices, one merging a with b and another merging b with c .

2.2 Greatest compatibility exactly at transitivity

For any reflexive symmetric relation C , the formal development proves:

$$\boxed{\text{a greatest equivalence } E \subseteq C \text{ exists} \iff C \text{ is transitive}}$$

The forward direction is worth spelling out. Every compatible pair $(s, t) \in C$ generates a two-point equivalence—identity plus the pair in both directions. If a greatest compatible equivalence E exists, it must contain each such two-point kernel. Hence $C \subseteq E$. By compatibility, $E \subseteq C$, so $E = C$. Since E is an equivalence, C is transitive. Conversely, if the reflexive symmetric relation C is transitive, then C itself is an equivalence and trivially the greatest compatible one.

This theorem does not say that an arbitrary tolerance has a canonical “best approximation” by equivalences. It says the opposite: the greatest candidate exists only when the tolerance was already transitive.

3. The mediator-hole obstruction

The smallest obstruction has three states and one load-bearing query. Let a, b, c be pairwise distinct and choose $w \in W$ such that

$$w \in A_a \cap A_c, \quad w \notin A_b.$$

Call this a **mediator hole**: the endpoints share a query that the proposed middle state cannot answer.

Choose a decision profile for which a and c disagree at w . All other jointly admitted decisions may be set equal. Then:

- aC_db , because b does not admit w and no shared query exposes a disagreement;

- $bC_d c$ for the same reason;
- $\neg aC_d c$, because both endpoints admit w and disagree there.

The formal necessity proof uses an especially economical adversarial profile:

$$d(s, u) \iff (s = c \wedge u = w).$$

At the missing coordinate, only the right endpoint receives the distinguished truth value. The profile is not claimed to arise from every operational transition system. It witnesses the exact quantifier in the static theorem: failure of the domain condition defeats transitivity for *some* permitted decision assignment.

The obstruction also explains why generic overlap slogans are too weak. It may remain present when:

- every two domains intersect;
- all domains have a nonempty global intersection;
- the domains are nested or laminar.

The **NestedFailure** fixture realizes all three conditions while retaining an endpoint-shared query absent from the middle domain. Nonempty overlap does not ensure mediation.

4. Semantic repair: preserve availability

The first repair treats future availability as part of the semantics rather than discarding it.

The native formal model supplies a deterministic execution of finite words. Let $\text{Adm}(s, w)$ mean that word w is admissible from state s , let $s \cdot w$ be the resulting state, and let D be the governed hazard or decision predicate. Define

$$s \sim_F t \iff \forall w, \begin{cases} \text{Adm}(s, w) \leftrightarrow \text{Adm}(t, w), \\ \text{Adm}(s, w) \implies (D(s \cdot w) \leftrightarrow D(t \cdot w)). \end{cases}$$

Because admissibility domains are required to agree, the problematic asymmetric middle case is unavailable. The relation is reflexive, symmetric, and transitive. It therefore defines a quotient map

$$q : S \rightarrow S / \sim_F.$$

The formal definition is operational: an observation $O : S \rightarrow Y$ is future-sufficient when any two states it identifies are future-equivalent,

$$\text{FutureSufficient}(O) :\iff \forall s, t, O(s) = O(t) \implies s \sim_F t.$$

Because equality in the quotient represents $\sim_F$, this definition has the following quotient-kernel characterization:

$$\text{FutureSufficient}(O) \iff \ker O \subseteq \ker q$$

This kernel-refinement property is the precise sense in which the quotient is the coarsest available abstraction for the declared future semantics. The formal result does not require a total factor map from arbitrary values outside the range of O , and it does not state a cardinality-minimality theorem.

The construction is analogous in spirit to Myhill–Nerode future equivalence [9], but the paper does not identify its deterministic hazard model with language recognition, automata minimization, trace equivalence, or bisimulation.

5. Structural repair: domain mediation

Suppose instead that the architecture intentionally ignores differences in availability. The relation remains C_d , and canonicity must come from a property of the domains.

Define **distinct-domain mediation** by

$$\forall a, b, c \in S, \quad (a, b, c \text{ pairwise distinct}) \implies A_a \cap A_c \subseteq A_b.$$

Repeated-state cases are excluded because transitivity is already immediate when $a = b$, $b = c$, or $a = c$. The condition says that every query shared by two distinct endpoints is visible to any third state that could serve as a genuine middle element.

Under this hypothesis, the transitivity proof is direct. Assume aC_db and bC_dc . Take $w \in A_a \cap A_c$. Domain mediation gives $w \in A_b$. The first compatibility premise yields

$$d_a(w) \leftrightarrow d_b(w),$$

and the second yields

$$d_b(w) \leftrightarrow d_c(w).$$

Therefore $d_a(w) \leftrightarrow d_c(w)$, so aC_dc .

The proof assumes no finiteness, decidable equality, quotient construction, or choice principle.

6. Domain mediation is Δ -system geometry

Distinct-domain mediation has a standard set-family form. Define a **common-root family** by the existence of a predicate $K \subseteq W$ such that

$$A_s \cap A_t = K \quad \text{for every } s \neq t.$$

This is exactly a Δ -system. Writing $P_s = A_s \setminus K$ gives the familiar decomposition

$$A_s = K \cup P_s, \quad P_s \cap P_t = \emptyset \quad (s \neq t).$$

No finiteness assumption is present. “Sunflower” is common informal terminology, but “ Δ -system/common-root family” avoids suggesting that the continuation languages are finite.

The formal development proves:

$$\boxed{\text{DistinctDomainMediator}(A) \iff \text{HasCommonCore}(A)}$$

One direction is immediate: if $A_a \cap A_c = K$ for distinct endpoints, then $K \subseteq A_b$ for every state b , so each endpoint-common query is admitted by the middle.

For the other direction, choose the common core

$$K = \bigcap_{s \in S} A_s.$$

The universal core is contained in every pairwise intersection. Conversely, if $w \in A_a \cap A_c$ for distinct a, c , domain mediation places w in every other state domain; the endpoint memberships already cover a and c . Hence $w \in K$. All distinct pairwise intersections therefore equal K .

The statement remains well formed for small or empty state types. Its distinct-pair clauses become appropriately vacuous rather than forcing all domains equal through the identity $A_s \cap A_s = A_s$.

7. Main characterization

Let **UniformCompatibilityTransitive** A mean that for every proposition-valued decision profile d , the relation C_d is transitive. The main theorem is:

$\begin{aligned} &\text{UniformCompatibilityTransitive}(A) \\ &\iff \text{DistinctDomainMediator}(A) \\ &\iff \text{HasCommonCore}(A). \end{aligned}$

The forward implication is the hostile-labeling argument from Section 3: if a mediator hole exists, one decision profile exposes it and violates transitivity. The reverse implication is the transport-through-the-middle argument from Section 5. The equivalence with a common root is Section 6.

This locates a sharp boundary for a particular question:

Which domain families guarantee transitivity independently of the static decision values assigned on those domains?

It does not answer the different fixed-model question:

For this one domain family and this one decision profile, is compatibility transitive?

The fixed-model question may have a positive answer outside the Δ -system class. For example, if every exposed decision is false, every pair is compatible no matter how the domains overlap. The universal relation is then the greatest compatible equivalence. Bad geometry creates the capacity for hidden endpoint disagreement; a particular labeling may fail to activate that capacity.

We call the two cases:

- **structural canonicity**: the domains force transitivity for every profile;
- **accidental or fixed-profile canonicity**: one profile happens to be transitive despite a mediator hole.

The theorem characterizes only the first.

8. Strictness and hostile fixtures

The formal fixtures delimit what the theorem does and does not require.

8.1 Private petals: equal domains are unnecessary

Take three states and four queries:

$$W = \{k, p_a, p_b, p_c\}.$$

Let

$$A_a = \{k, p_a\}, \quad A_b = \{k, p_b\}, \quad A_c = \{k, p_c\}.$$

All distinct intersections equal $K = \{k\}$, but the domains are not equal. The uniform theorem supplies transitivity for every decision profile, and the common-domain relation itself is the greatest compatible equivalence.

In the native word model, the **PrivatePetals** fixture uses the empty word as the common root and private self-loop controls as petals. It proves both that the new condition is strictly weaker than globally equal continuation domains and that two states may be common-domain compatible while failing the stronger future equivalence because their availability domains differ.

Thus canonicity of the weaker decision relation does not require preservation of complete future availability information.

8.2 Nested failure: pleasant overlap is insufficient

Let

$$A_a = A_c = \{u, v\}, \quad A_b = \{u\}.$$

The domains are nested. Every pair overlaps. The global intersection $\{u\}$ is nonempty. Yet $v \in A_a \cap A_c$ and $v \notin A_b$. Give a and c different decisions at v while preserving agreement at u . Then

$$aC_db, \quad bC_dc, \quad \neg aC_dc.$$

Laminarity, pairwise nonempty overlap, and nonempty global intersection do not remove mediator holes.

8.3 Constant decisions: geometry is not fixed-model necessity

Keep the nested domains but assign the same proposition at every admitted query. Compatibility becomes universal and transitive. This fixture prevents a misreading of the main theorem: common-root geometry is necessary for the uniform domain-only guarantee, not for every individual assignment.

9. Architecture interpretation

The formal result supports the following bounded design rule:

If an architecture defines compatibility as agreement on everything both sides can answer, varying answerability domains can make compatibility nontransitive. A canonical abstraction requires either preserving answerability in the semantics or enforcing domain geometry that prevents mediator holes.

The two repairs preserve different information.

Architecture choice	Preserves	Canonicity basis	Cost
Availability-sensitive future equivalence	domain availability and governed decisions	equivalence by construction	distinguishes states whose answers match but capabilities differ
Common-domain compatibility under Δ -system domains	governed decisions on shared queries only	structural transitivity	restricts which queries may be shared outside the common core

Neither route dominates the other without application semantics. If the fact that a query is unavailable matters, suppressing that difference is already the wrong model. If availability is intentionally outside the decision semantics, the Δ -system condition identifies one exact way to retain canonicity without forcing every domain to be equal.

9.1 A minimal monitor example

Consider a fleet health API that reports proposition-valued predicates. Every agent reports the core predicate `service_ready`; optional hardware plugins report additional predicates.

Three agent states have domains:

A: {`service_ready`, `disk_fault`}

B: {`service_ready`}

C: {`service_ready`, `disk_fault`}

All three report `service_ready = true`. Agent A reports `disk_fault = false`; agent C reports `disk_fault = true`. If the grouping rule is “agree on all predicates both expose,” then A is compatible with B and B with C, while A is incompatible with C.

There are two honest repairs.

1. Include the exposed-predicate set in the observation, so B is not equivalent to A or C merely because it lacks the disputed predicate.
2. If the architecture insists on ignoring capability differences, prevent two agents from sharing an optional predicate unless that predicate belongs to the common core exposed by every agent. All non-core predicates must be private to one domain.

The second policy is exact for the static model and often operationally restrictive. The example is therefore an illustration, not a recommendation that real monitoring fleets should adopt private-petal schemas. Dynamic capabilities, noisy values, multivalued outputs, authorization checks, and temporal sampling are outside the theorem.

10. Related work

10.1 Partial functions and compatibility

Borlido and McLean study algebras represented by partial functions and use the relation “agree on the intersection of their domains” as the canonical compatibility relation [1]. Their representation theorem demonstrates that this is an established and general object. The present result does not compete with their algebraic theory; it asks a narrower structural question about uniform transitivity when domains are frozen and values vary.

10.2 Tolerance relations

Tolerance theory studies reflexive symmetric relations under many historical names, including compatibility and similarity [2,3]. Its maximal blocks are maximal cliques and need not form a quotient partition. This supplies the general relation-theoretic background for the common-domain relation. The paper’s greatest-compatible result is an elementary kernel-order fact, not a new theory of tolerances.

10.3 Incompletely specified machines

Paull and Unger’s classical work on incompletely specified sequential machines replaces complete-machine equivalence classes with compatibility classes and covers [4]. Modern model-based testing continues to confront compatible states in partial specifications [5]. This is the closest operational precedent: partial behavior complicates state merging because compatibility need not be an equivalence. Machine compatibility may impose transition closure or implementation constraints not present in the static profile theorem, so the two results should not be identified.

10.4 Partial words

In partial-word combinatorics, wildcard positions yield a compatibility relation that is expressly not an equivalence relation [6]. A defined symbol, a hole, and a conflicting defined symbol realize the same local

shape as the mediator-hole example. The domain-family uniform characterization is not the object studied in that literature.

10.5 Δ -systems and partial-function amalgamation

Δ -systems are classical set families with a common pairwise intersection root [7]. In forcing, the Δ -system lemma is routinely applied to domains of finite partial functions; after restricting to functions that agree on the root, their unions are compatible. Sánchez Terraf formalizes the Δ -system lemma and its countable-chain-condition application in Isabelle/ZF [8]. Those arguments use common-root domains to extract pairwise-compatible subfamilies. The theorem here instead characterizes when the pairwise compatibility relation is transitive under every labeling. The ingredients are the same; the quantifier and architecture interpretation differ.

10.6 Future quotients and agreement testing

The availability-sensitive quotient follows a Myhill–Nerode-shaped pattern: states are identified by their behavior under all future words [9]. The formal result is limited to its declared deterministic continuation and hazard semantics.

Agreement testing asks when local functions that agree on overlaps are close to one global function [10]. That is a richer approximate local-to-global problem. This paper asks neither for approximate decoding nor for a global amalgam; it asks when pairwise agreement itself is transitive.

No exact prior publication of the uniformly quantified Δ -system characterization was located in the bounded review that produced this draft. Because the proof is an immediate mediator-hole argument from standard definitions, the paper makes no claim of new combinatorics on that basis.

11. Lean formalization and artifact boundary

The formal development separates four layers:

1. static proposition-valued profiles and their domains;
2. a deterministic finite-word continuation model;
3. relation/kernel results about compatible observations;
4. finite hostile and strictness fixtures.

The central static declarations are:

```
ProfileCompatible
DistinctDomainMediator
HasCommonCore
UniformCompatibilityTransitive
commonCore_iff_domainMediator
uniformCompatibilityTransitive_iff_domainMediator
```

The continuation and canonicity surface includes:

```
CommonCompatible
PartialFutureSufficient
greatestCompatible_exists_iff_transitive
futureSufficient_iff_kernel_refines
commonCompatible_trans_of_domainMediator
greatestCommonCompatible_exists_of_domainMediator
```

The focused campaign gate compiled 40 declarations in the domain-overlap layer and reran 31 partial-domain, 23 continuation-quotient, 16 continuation-hazard, and 57 compositional-admissibility regression declarations. It found no `sorry`, `sorryAx`, `admit`, or local axiom declarations. The principal structural and sharpness theorems were axiom-free; inherited quotient and finite-fixture declarations account for the broader reported footprint of `Classical.choice`, `Quot.sound`, and `propext`.

The current paper directory is not yet a standalone formal artifact. The verified development revision and toolchain are recorded in `FORMAL-ARTIFACT.md`; a minimal portable extraction and fresh-clone replay are release tasks. Until those are complete, this manuscript should be described as a draft backed by verified development evidence, not as an independently replayable package.

12. Limitations

The theorem has a deliberately narrow scope.

1. **Static profile quantifier.** Uniform sharpness ranges over arbitrary proposition-valued static profiles. It does not prove that every adversarial profile is realizable by a given transition system.
2. **Fixed-model necessity is false.** A particular decision assignment may be transitive outside the Δ -system class.
3. **No cardinality result.** Greatest equivalence under relation inclusion is not a theorem about the smallest observation codomain or minimum number of monitor states.
4. **No behavioral equivalence.** Common compatibility and the stronger future relation are not claimed to be bisimulation, trace equivalence, or complete operational semantics.
5. **No dynamic or probabilistic domains.** Availability is frozen, and decisions are proposition-valued. Noise, time variation, and partial trust are absent.
6. **No production validation.** The monitor example is an exact model illustration, not evidence about a deployed API.
7. **No novelty claim for the ingredients.** Partial-function compatibility, tolerance nontransitivity, incomplete-machine covers, and Δ -systems are established.
8. **No empirical promotion.** The theorem does not repair or validate any neural-representation experiment.

The nearest unpursued mathematical question is fixed-profile transitivity outside the Δ -system class. Mediator holes create potential violations, but a particular labeling may neutralize them. The present work does not establish whether those exceptions admit a clean forbidden-pattern theory or only a constraint catalog.

13. Conclusion

Agreement on common domains is a natural relation for partial observations. It is not, by itself, a safe equality surrogate. A state that lacks the query on which two endpoints disagree can mediate two compatible comparisons while the endpoints remain incompatible.

There are two clean ways to recover canonical abstraction. The semantic repair preserves availability and outcomes together, yielding a future equivalence and a canonical quotient. The structural repair keeps the weaker, domain-ignoring semantics but constrains the domains. Uniformly over static decision profiles, the exact condition is Δ -system/common-root geometry.

The practical lesson is modest and precise:

Pairwise agreement over shared answerability does not automatically compose. If answerability is erased, canonicity requires a structural reason.

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